

linezolid-amr

# linezolid-amr

Integrated AMR profiling + 23S rRNA  
linezolid-resistance frequency analysis

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**User Guide** · v0.2.0 · 2026-08-04

*S. aureus* · *E. faecalis* · *E. faecium* · *S. pneumoniae*

In-house MLST (PubMLST) → AMRFinderPlus → species-aware 23S read pileup. One command, one combined report, two summary CSVs (wide Kleborate-style + long per-feature), organism auto-detection, folder batch mode.

# 1 · Quick install

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## From bioconda recommended

```
conda create -n linezolid-amr -c bioconda -c conda-forge linezolid-amr
conda activate linezolid-amr
```

## From source

```
git clone https://github.com/iowa69/linezolid-amr && cd linezolid-amr
conda create -n linezolid-amr -c bioconda -c conda-forge \
    python=3.11 ncbi-amrfinderplus minimap2 samtools bcftools mlst
conda activate linezolid-amr
pip install -e .
```

## One-time setup (the tool auto-fetches if missing)

```
amrfinder -u                # ~150 MB
linezolid-amr fetch-references # ~120 KB total
```

## 2 · Pipeline overview

| Step                    | Tool                                       | Output                                                                      |
|-------------------------|--------------------------------------------|-----------------------------------------------------------------------------|
| 1. MLST & organism      | in-house blastn vs bundled PubMLST schemes | Sequence Type + auto-detected organism                                      |
| 2. AMR profile          | AMRFinderPlus                              | All AMR / virulence / stress genes & mutations                              |
| 3. 23S heteroresistance | minimap2 + pysam + bcftools                | Allele frequency at 18 curated LZD positions, per-allele evidence, full VCF |

**Why read-level for 23S?** *S. aureus* and *Enterococcus* spp. carry 4–6 copies of the 23S rRNA operon. A G2576T mutation present in only 1–3 copies produces a 15–60 % allele-fraction heteroresistant signal — invisible to assembly-only callers (consensus base stays G). Reading the raw reads is the only way to catch it.

**Telling heteroresistance from artifact.** Because the signal is a *minority* allele it shares the frequency range with sequencing and alignment noise. Each candidate allele is therefore written to `<sample>.23S_lzd_evidence.tsv` with the evidence needed to separate the two: forward/reverse read counts and a Fisher exact strand-bias p-value (a real mutation is carried by the template and appears on both strands; an allele seen on one strand only is rejected), a Wilson 95 % confidence interval on the fraction, and the number of rrn operons that fraction implies. Each position also carries an **evidence tier** — substitutions only ever selected in archaea or mycobacteria are reported with their fractions but cannot drive a clinical positive unless you pass `--evidence-tier experimental`.

## 3 · Usage

### 3.1 · Single sample

```
linezolid-amr run \  
  -a sample.fasta \  
  -1 sample_R1.fq.gz -2 sample_R2.fq.gz \  
  -o results/sample
```

MLST infers the organism. Override with `-O Enterococcus_faecium`. A warning prints if your override disagrees with MLST.

### 3.2 · Folder mode (batch)

```
linezolid-amr folder -i input_dir/ -o results/
```

Drops every `*.fasta/*.fa/*.fas/*.fna` into a sample, pairs it with FASTQs named with the same basename. Recognised read suffixes:

| Pattern                        | Example                                              |
|--------------------------------|------------------------------------------------------|
| <code>_R1_001 / _R2_001</code> | <code>S1_R1_001.fastq.gz / S1_R2_001.fastq.gz</code> |
| <code>_R1 / _R2</code>         | <code>S1_R1.fq.gz / S1_R2.fq.gz</code>               |
| <code>_1 / _2</code>           | <code>S1_1.fastq.gz / S1_2.fastq.gz</code>           |

Produces two cohort-level CSVs at `results/`: `ALL_samples.summary_wide.csv` (one row per sample) and `ALL_samples.summary_long.csv` (one row per gene/mutation).

### 3.3 · All options

| Flag                         | Default          | Meaning                                          |
|------------------------------|------------------|--------------------------------------------------|
| <code>-a / --assembly</code> | —                | FASTA assembly (run only)                        |
| <code>-1 / --r1</code>       | —                | Forward FASTQ (run only)                         |
| <code>-2 / --r2</code>       | —                | Reverse FASTQ (optional)                         |
| <code>-i / --input</code>    | —                | Input folder (folder only)                       |
| <code>-o / --outdir</code>   | —                | Output directory                                 |
| <code>-s / --sample</code>   | assembly<br>stem | Sample name                                      |
| <code>-O / --organism</code> | auto (MLST)      | Override MLST organism inference                 |
| <code>-t / --threads</code>  | all CPUs         | Parallel threads                                 |
| <code>--plus</code>          | off              | AMRFinderPlus stress/virulence/biocide search    |
| <code>--min-af</code>        | 0.15             | Min 23S alt-allele frequency for a positive call |
| <code>--min-depth</code>     | 20               | Min depth at a 23S position                      |
| <code>--min-baseq</code>     | 13               | Discard bases below this Phred quality           |

|                                                |            |                                                                    |
|------------------------------------------------|------------|--------------------------------------------------------------------|
| <code>--min-mapq</code>                        | 20         | Discard reads below this mapping quality                           |
| <code>--min-alt-reads</code>                   | 3          | Min reads supporting an allele before it can be called             |
| <code>--strand-bias-p</code>                   | 1e-3       | Fisher exact p below which strand distribution counts as biased    |
| <code>--no-strand-filter</code>                | off        | Report strand-biased alleles as positive calls (pre-0.2 behaviour) |
| <code>--evidence-tier</code>                   | associated | Weakest evidence tier allowed to drive a POSITIVE call             |
| <code>--reads-only</code>                      | off        | (folder) process every R1/R2 pair without assemblies               |
| <code>--skip-amrfinder / --skip-rrna23s</code> | —          | Skip one step                                                      |

## 4 · Output structure

```
results/sample/
├── amrfinder/amrfinder.tsv          # all AMR / virulence / stress hits
├── rrna23s/
│   ├── sample.23S.bam (+ .bai)     # sorted, indexed alignment
│   ├── sample.23S.vcf.gz (+ .csi)  # all 23S variants
│   └── sample.23S_lzd_pileup.tsv    # AF at canonical LZD sites
├── sample.linezolid_amr.json        # combined machine-readable report
├── sample.linezolid_amr.txt         # plain-text summary
├── sample.summary_wide.csv          # 1 row · AMR class buckets as cols
└── sample.summary_long.csv          # 1 row per gene/mutation
```

In **folder** mode each sample sits in its own subfolder, plus the two cohort CSVs at the root of `outdir/`.

### summary\_wide.csv columns (fixed prefix)

| Column                     | Meaning                                                           |
|----------------------------|-------------------------------------------------------------------|
| sample                     | sample name                                                       |
| organism                   | AMRFinderPlus organism used                                       |
| mlst_scheme / ST           | MLST scheme + sequence type                                       |
| linezolid_call             | <code>POS</code> if any LZD signal, else <code>neg</code>         |
| lzd_23S_mutations          | e.g. <code>G2576T:0.6647(173)</code> — pos:af(depth)              |
| lzd_genes                  | e.g. <code>cfr(D);optrA;23S_G2576T</code>                         |
| AMR_<class>                | one column per AMRFinderPlus class — <code>gene(cov/ident)</code> |
| VIRULENCE / STRESS_<class> | only present with <code>--plus</code>                             |

# 5 · Worked example — Enterococcus faecium VRE LZD-R

```
$ linezolid-amr run -a test_assembly.fasta -1 test_R1.fq.gz -2 test_R2.fq.gz \
-o results/test_efaecium -s test -t 8

=== test ===
>> MLST / organism inference...
  organism: Enterococcus_faecium   MLST scheme: efaecium   ST: 17
>> Running AMRFinderPlus...
  17 hits, 3 linezolid-relevant
>> Running 23S rRNA analysis...
  18 positions; 1 with resistance allele

Linezolid resistance call: POSITIVE
```

## Linezolid signal

| Source                     | Detection                                    |
|----------------------------|----------------------------------------------|
| AMRFinderPlus              | 23S_G2576T , cfr(D) , optrA                  |
| 23S pileup at E. coli 2576 | G=58, T=115 (AF=0.665) — ~4/6 operoni mutati |

## summary\_wide.csv excerpt

```
sample,organism,mlst_scheme,ST,linezolid_call,lzd_23S_mutations,lzd_genes,AMR_OXAZOLIDINONE,AMR_GLYCOPEPTIDE,AMR_BETA-LACTAM,...
test,Enterococcus_faecium,efaecium,17,POS,G2576T:0.6647(173),"cfr(D);optrA;23S_G2576T",
"cfr(D)(100/100.0);optrA(100/99.7);23S_G2576T(100/100.0)","vanA(100/99.7);vanH-
A(100/100.0);vanX-A(72/100.0);vanY-A(100/99.7);vanZ-
A(100/100.0)","pbp5_E629V(100/95.4);pbp5_M485A(100/95.4)",...
```

**Heteroresistance read-out.** E. faecium has 6 rrn operons. 0.665 ≈ 4/6 mutated copies — a typical clinical heteroresistance pattern. The tool quantifies it exactly; conventional callers would just report *G2576T present* without the operon fraction.

## 6 · References

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All canonical 23S positions in `loci.json` carry a verified PMID. Full bibliography in `CITATION.cff`.

| PMID     | Reference                    | Role                                |
|----------|------------------------------|-------------------------------------|
| 10556031 | Kloss et al. 1999, JMB       | Foundational 23S mutational mapping |
| 11476839 | Tsiodras et al. 2001, Lancet | First clinical G2576T description   |
| 22143525 | Long & Vester 2012, AAC      | Comprehensive resistance review     |
| 16801432 | Long et al. 2006, AAC        | Cfr methyltransferase mechanism     |
| 25977397 | Wang et al. 2015, JAC        | optrA discovery                     |
| 29635422 | Antonelli et al. 2018, JAC   | poxTA discovery                     |